

Splunk Alert Detection & Notification

Proof of Concept — SOC Internship, Splunk Enterprise Evaluation

Extracted from: Splunk Enterprise — Hands-On Evaluation Report | September 17, 2026

1. Objective

An alert is a saved search that Splunk keeps re-running on a schedule, firing a notification the moment its condition is met — turning a one-time search into always-on monitoring. This document is a focused proof of concept for the alerting capability of Splunk Enterprise, covering three things end-to-end: the alerts that were configured, confirmation that each one actually runs and evaluates on schedule (not just saved, but live), and confirmation that a real notification is delivered when a condition fires.

2. Alerts Configured

Five alerts were built against the SSH connection log, each enabled and scheduled to run automatically. All five appear as "Enabled" in the Alerts listing below.

Import favorites | reset password tool

splunk>Enterprise

Alerts

Alerts set a condition that triggers an action, such as sending an email that contains the results of the triggering search to a list of people. Click the name to view the alert. Open the alert in Search to refine the parameters.

5 Alerts of 1 pages

Title ↑	Actions	Next scheduled time ↓	Owner ↓	App ↓	Sharing ↓	Status ↓
> Accepted public key	Open in search Edit	Sep 16, 2026 4:20:00 PM	admin7	search	Private	✓ Enabled
> Failed Login	Open in search Edit	Sep 16, 2026 4:20:00 PM	admin7	search	Private	✓ Enabled
> Multiple attempts	Open in search Edit	Sep 16, 2026 4:20:00 PM	admin7	search	Private	✓ Enabled
> Multiple failed	Open in search Edit	Sep 16, 2026 4:20:00 PM	admin7	search	Private	✓ Enabled
> Opened	Open in search Edit	Sep 16, 2026 4:20:00 PM	admin7	search	Private	✓ Enabled

Figure 1 — Five alerts configured and enabled: Accepted public key, Failed Login, Multiple attempts, Multiple failed, and Opened.

2.1 What Each Alert Means

Alert	Fires When	Why It Matters
Accepted public key	An SSH log entry records a successful key-based login (an “Accepted publickey” event).	Tracks every successful key-based access to a server — the events you most want visibility into, since a real breach ends in a successful login, not just an attempt.
Failed Login	A single failed SSH authentication attempt is recorded.	The most basic building block for brute-force detection — one failed attempt on its own is often normal, but the raw signal feeds the two threshold-based alerts below.
Multiple attempts	A single source IP generates more authentication attempts (successes and failures combined) than the defined threshold within the search window.	Flags any source that is unusually active against SSH, regardless of whether those attempts succeeded — a broader net than looking at failures alone.
Multiple failed	A single source IP generates more failed authentication attempts than the defined threshold within the search window.	The core brute-force detection alert — a high volume of failures from one source is the clearest signature of password guessing.
Opened	A new SSH connection/session is opened against a monitored host.	Baseline visibility into every new session being established, successful or not — a general activity trip-wire independent of authentication outcome.

Trigger conditions above are summarized from each alert's name and standard SSH log semantics; refer to the saved search definition in Splunk for the exact SPL condition and threshold value used by each alert.

3. Confirming Alerts Actually Trigger

Being listed as “Enabled” only confirms an alert is configured — it does not confirm the alert is actually evaluating on schedule. To verify that, the job history for the “Failed Login” saved search was checked directly.

The screenshot shows the Splunk Job History interface for a search labeled "Failed Login". The interface includes filters for Owner (All owners), App, and Status (All statuses). It shows 10 results as of Sep 10, 2026. The table lists jobs with columns for Job, Owner, App, Events, Size, Created at, Expires, Runtime, and Status. The jobs are scheduled and completed on Sep 10, 2026, and Sep 11, 2026.

Job	Owner	App	Events	Size	Created at	Expires	Runtime	Status
Failed Login	admin7	search	0	280 KB	Sep 10, 2026 9:58:20 AM	Sep 10, 2026 10:54:25 AM	00:54:05	Running (real-time)
Failed Login	admin7	search	1	84 KB	Sep 10, 2026 9:58:20 AM	Sep 11, 2026 9:59:48 AM	00:01:28	Done
Failed Login	admin7	search	1	92 KB	Sep 10, 2026 9:58:20 AM	Sep 11, 2026 9:59:58 AM	00:01:38	Done
Failed Login	admin7	search	1	236 KB	Sep 10, 2026 9:58:20 AM	Sep 11, 2026 10:40:14 AM	00:41:54	Done
Failed Login	admin7	search	1	236 KB	Sep 10, 2026 9:58:20 AM	Sep 11, 2026 10:40:17 AM	00:41:57	Done
Failed Login	admin7	search	1	236 KB	Sep 10, 2026 9:58:20 AM	Sep 11, 2026 10:40:20 AM	00:41:60	Done
Failed Login	admin7	search	1	236 KB	Sep 10, 2026 9:58:20 AM	Sep 11, 2026 10:40:22 AM	00:42:02	Done
Failed Login	admin7	search	1	236 KB	Sep 10, 2026 9:58:20 AM	Sep 11, 2026 10:40:27 AM	00:42:07	Done
Failed Login	admin7	search	1	252 KB	Sep 10, 2026 9:58:20 AM	Sep 11, 2026 10:44:53 AM	00:46:33	Done
Failed Login	admin7	search	1	252 KB	Sep 10, 2026 9:58:20 AM	Sep 11, 2026 10:45:01 AM	00:46:41	Done

Figure 2 — Job history for the “Failed Login” saved search, showing it has run repeatedly on schedule (multiple completed jobs with event counts and runtimes) — proof the alert is live, not just configured.

4. Confirming Alert Delivery (Email)

The final link in the chain is delivery: does a real notification actually reach an inbox when a condition fires? Two separate alert conditions were triggered and their email notifications captured below, confirming the full loop — condition defined, evaluated on schedule, and delivered — works end-to-end, not just in configuration.

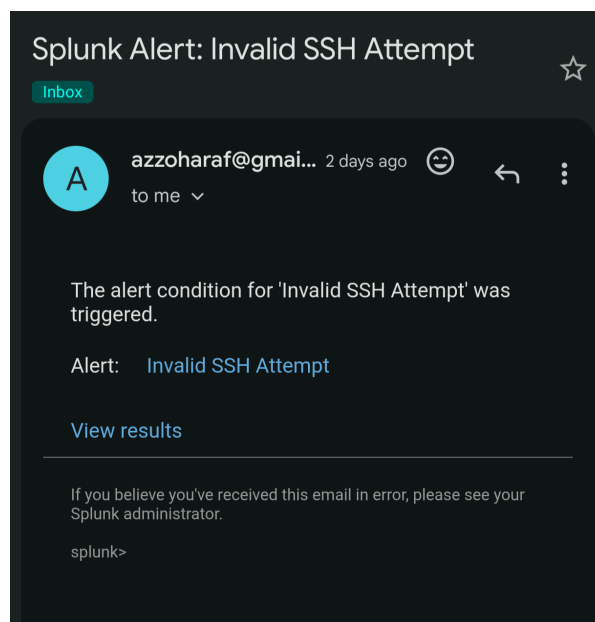


Figure 3 — Email notification received for the “Invalid SSH Attempt” alert condition being triggered.

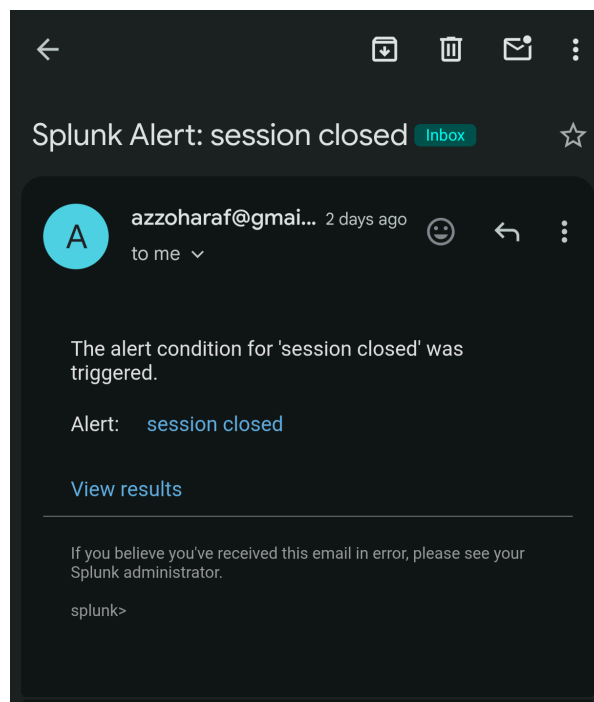


Figure 4 — Email notification received for the “session closed” alert condition being triggered.

5. Alternative Alert Delivery Method

In addition to Splunk's built-in scheduled-alert email action shown above, an alert notification was also configured and confirmed using a separate delivery method outside Splunk's standard trigger path. *[Evidence for this method to be added — see note below.]*

This section is a placeholder pending the additional screenshots for the alternate mail-trigger setup — once supplied, this section should describe how that method was configured (e.g. a webhook, script, or third-party integration), how it differs from Splunk's native alert action, and include the corresponding evidence image(s) with the same figure/caption format used above.

6. Summary

Together, these confirm the full alerting loop end-to-end: a condition is defined, Splunk evaluates it on schedule, and a real notification reaches the analyst's inbox when it fires — not just that the alert exists, but that it works. Five alerts covering successful logins, failed logins, and threshold-based brute-force detection were configured against the SSH connection log, confirmed live via job history, and confirmed delivering real notifications on trigger.